Week 2 Lesson 2: Similarities of Computer Ethics with Other Ethics, Professional Ethics & General Ethics

Learning Objectives:

By the end of this lecture, students will be able to:

- Understand what computer ethics is and how it relates to other forms of ethics.
- Identify similarities between computer ethics, general ethics, and professional ethics.
- Evaluate ethical issues related to software engineering.
- Analyze scenarios and make ethical decisions in professional computing practice.

Introduction to Computer Ethics

Computer Ethics is the branch of ethics that examines moral dilemmas and responsibilities in the use and development of computers and software.

Definition by James H. Moor (1985): "Computer ethics is the analysis of the nature and social impact of computer technology and the corresponding formulation and justification of policies for the ethical use of such technology."

Key Areas in Computer Ethics:

- Data privacy and security
- Software piracy
- Intellectual property
- Responsibility of software engineers
- Use of AI, robotics, and automation

General Ethics vs. Computer Ethics

General Ethics	Computer Ethics
Concerned with right/wrong behavior in society	Deals with moral decisions involving technology
Based on universal moral values (e.g., honesty, fairness)	Applies traditional values to new tech contexts
Used in all fields of life	Specific to computer and information systems

Similarities:

- Both require **honesty**, **fairness**, **respect**, and **accountability**

- Both focus on minimizing **harm**
- Involve decision-making that affects others

Example: In general ethics, stealing is wrong. In computer ethics, copying software without a license (piracy) is stealing.

Professional Ethics vs. Computer Ethics

Professional Ethics is a set of standards that guide professionals in their conduct within a profession.

Similarities:

- Both define **acceptable behavior** in a professional context
- Both require **responsibility and accountability**
- Both emphasize **confidentiality, competence, and integrity**

Codes of Conducts:

- **ACM Code of Ethics**
- **IEEE Code of Ethics**
- **BCS Code of Conduct**

Example for Software Engineers:

Writing secure code is a matter of professional ethics. Ignoring known vulnerabilities is unethical and unprofessional.

Similarities Between Ethics Types

Aspect	General Ethics	Professional Ethics	Computer Ethics
Honesty	Truthful behavior	Accurate reporting	No falsifying data or code
Fairness	Treating others equally	No discrimination	No algorithmic bias
Responsibility	Accepting consequences	Accountable for tasks	Accountable for systems' outcomes
Respect for Others	Dignity, privacy	Client confidentiality	Data privacy and user consent
Integrity	Acting consistently with morals	Professional transparency	No plagiarism or software piracy

Common Ethical Principles in All Domains

Principle	Description	Example in Software Engineering
Do No Harm	Avoid causing harm to others	Avoid releasing buggy software

Honesty	Tell the truth, don't mislead	Clearly explain software limits
Fairness	Avoid discrimination or bias	Build inclusive software
Respect for Rights	Respect user privacy and property	Do not steal or misuse data
Accountability	Be responsible for your actions	Take ownership of code issues

Scenario-Based Examples

Scenario 1: Software Piracy

A student downloads a paid software using a crack version to complete an assignment.

- **General Ethics**: This is theft.
- **Computer Ethics**: It violates intellectual property laws.
- **Professional Ethics**: A future professional must learn to respect legal software use.

Scenario 2: Biased AI Algorithm

A developer designs a facial recognition system that performs poorly on women and minorities.

- **General Ethics**: It's unjust and discriminatory.
- **Computer Ethics**: Introduces harmful bias into society.
- **Professional Ethics**: The engineer must fix the issue, test across diverse datasets.

Scenario 3: Data Breach Cover-Up

A team discovers a serious data leak but chooses to hide it to protect the company image.

- **General Ethics**: It's dishonest.
- **Computer Ethics**: Violates privacy and trust.
- **Professional Ethics**: Ethical conduct demands disclosure and fixing the problem.

Relevant Laws and Regulations

- **Bangladesh ICT Act 2006** – Regulates electronic crimes and responsibilities.
- **Digital Security Act 2018 (BD)** – Penalizes illegal access, data breaches, defamation.
- **GDPR (EU)** – Governs data privacy and user consent (globally influential).
- **DMCA (USA)** – Digital content protection and copyright.

Importance for Software Engineers

- Software Engineers often control powerful systems (banking, medical devices, security).
- Their choices can have **life-changing consequences**.
- Professional and computer ethics ensure:
  - User trust
  - System security

- Long-term sustainability

Summary

Ethics Type	Scope	Example in Computing
General Ethics	All humans	Don't lie, don't steal
Professional Ethics	People in a profession	Respect confidentiality
Computer Ethics	Tech-specific scenarios	Avoid spreading malware

Sample Exam Questions

Short Question:

1. Define computer ethics and explain how it relates to general ethics.
2. What is the role of professional ethics in software engineering?
3. Provide one example where all three ethics (general, professional, computer) overlap.

References

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